

VPA IB DP CS — Complete IA Handbook

VPA IB DP Computer Science — Complete IA Handbook

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Weight: 30% SL / 20% HL | Word Count: No official maximum (quality over quantity)

Part 1: Understanding the IA

What is the IA?

The Internal Assessment is a software development project that solves a real problem for a real client. It is not a hypothetical exercise — the client must exist, the problem must be genuine, and the solution must be implemented and evaluated.

What Makes a Good IA?

Feature	Good IA	Poor IA
Client	Real person with a real organisational role; documented contact	"My friend" with no clear role; invented client
Problem	Specific, genuine, currently causing difficulty	Vague, already solved, purely hypothetical
Solution	Software that addresses the problem directly	Generic website/app with no clear problem-solution link
Complexity	Uses at least two complex techniques from the IB list	Simple calculator or basic form with no complex techniques
Evidence	Client emails, meeting notes, screenshots, video	No evidence of client involvement

Part 2: Criteria Breakdown

Criterion A: Planning (6 marks)

A.1 — The Client and Problem

Introduce your client, their organisation, and their role. Quote the client describing the problem in their own words.

Exemplar Extract:

"Ms. Chen Li is the Operations Manager at VPA Shenzhen's cafeteria. She manages food orders for 400 students daily, tracking inventory, recording allergies, and generating weekly purchase reports. Currently, she uses a paper logbook and Excel spreadsheets, which leads to lost records, inability to track allergens in real time, and hours spent manually compiling reports each Friday. Ms. Chen stated: 'I spend more time doing paperwork than managing the kitchen. I need a system that tracks everything automatically.'"

A.2 — Rationale for Solution

Explain why a computational solution is appropriate. Consider alternatives (manual, commercial software) and justify your custom solution.

A.3 — Success Criteria (3–5 criteria)

Each criterion must be specific, measurable, and testable. Avoid vague language.

Weak Criterion	Strong Criterion
"The system should be easy to use"	"The system shall allow a new food item to be added in fewer than 60 seconds without training"
"The system should be fast"	"The system shall generate a weekly report in fewer than 5 seconds for 400 student records"
"The system should store data"	"The system shall store at least 1,000 student allergy records with persistent storage"

A.4 — Record of Tasks

A dated log showing every significant activity. Include evidence of client consultation.

Date	Task	Time	Client Consultation?
2025-09-10	Initial meeting with Ms. Chen to discuss cafeteria problems	30 min	Yes — meeting notes attached
2025-09-15	Drafted success criteria; sent to Ms. Chen for approval	45 min	Yes — email response received
2025-09-20	Researched Python GUI libraries (Tkinter, PyQt)	60 min	No

Criterion B: Design (6 marks)

B.1 — System Overview Diagram

Show the overall architecture: modules, data storage, user interface, data flow. Use tools like Lucidchart, Draw.io, or hand-drawn (scanned).

B.2 — Algorithms

Design your key algorithms in IB pseudocode before coding. Include:

Input and output specifications

Processing steps

Loops, conditions, and data structures

Complex techniques (e.g., file handling, searching, sorting)

Exemplar Algorithm (Student Record Search):

```
function searchStudent(studentID)
  open file "students.csv" for reading
  found ← false
  while not end of file do
    record ← read line from file
    if record.studentID = studentID then
      found ← true
    output record
  end if
end while
```

```

if found = false then
output "Student not found"
end if
close file
end function

```

B.3 — Test Plan

Test #	Success Criterion	Test Data	Expected Result	Type
1	Add item in <60s	Valid item name, price, category	Item saved; confirmation message	Normal
2	Generate weekly report in <5s	400 student records	Report displayed in 4.2 seconds	Boundary
3	Store 1,000 allergy records	1,001st record added	System rejects with error message	Erroneous

Criterion C: Development (12 marks)

C.1 — Complex Techniques

To score highly, demonstrate at least two complex techniques:

File handling (reading/writing structured data, CSV parsing)

Searching or sorting algorithms (binary search, merge sort)

Collections/arrays of records

Object-oriented programming (classes, inheritance, polymorphism)

Database connectivity (SQL queries, SQLite)

API integration (REST APIs, JSON parsing)

Recursive algorithms

Multi-threading or concurrency

C.2 — Code with Commentary

Include key code sections with inline commentary. Do not dump your entire codebase. Select the sections that demonstrate complexity and design.

Commentary Example:

```

// This function implements a binary search on the student array
// Binary search is chosen because the array is sorted by studentID
// and provides O(log n) efficiency rather than O(n) for linear search
function binarySearch(arr, target, low, high)
if low > high then
return -1 // Base case: element not found
end if
mid ← (low + high) / 2
if arr[mid].studentID = target then
return mid // Found at midpoint
else if arr[mid].studentID
return binarySearch(arr, target, mid + 1, high) // Search upper half

```

```

else

return binarySearch(arr, target, low, mid - 1) // Search lower half

end if

end function

```

C.3 — Video Evidence

A 5–7 minute narrated video demonstrating your solution. Requirements:

Show the program running from launch

Demonstrate each success criterion being met

Include narration explaining what is happening

Keep to 7 minutes maximum

Upload to YouTube (unlisted) or school server; provide link

Criterion D: Evaluation (4 marks)

D.1 — Testing Against Success Criteria

Show evidence for each criterion. Use screenshots, test data, and results.

Criterion	Test Method	Result	Evidence
Add item in <60s	Stopwatch timing of 5 trials	Average 42 seconds	Screenshot of timer + system
Report in <5s	Stopwatch timing with 400 records	4.2 seconds	Screenshot of output

D.2 — Client Feedback

Include direct client feedback. Examples:

Written evaluation form

Email response

Meeting notes

Audio/video recording

Exemplar Client Feedback:

"The system has significantly reduced my weekly paperwork from 3 hours to 15 minutes. I especially like the allergy alert feature — it has already prevented two potential incidents. However, I would like a Chinese language option in the future. Overall, I am very satisfied." — Ms. Chen Li, VPA Cafeteria Operations Manager

D.3 — Future Improvements

Identify at least two realistic improvements that could be made with more time or resources.

Multi-language support (Chinese interface)

Cloud synchronisation for multiple devices

Barcode scanning integration for faster inventory entry

Automated email alerts to suppliers when stock is low

Part 3: Exemplar Project Outline (Grade 7 Standard)

Project: VPA Library Management System

Client: Ms. Wang, School Librarian

Problem: Managing 5,000 books using a paper card system leads to lost books, untraceable loans, and no usage analytics.

Success Criteria:

Search for a book by title/author in under 3 seconds

Check out a book in under 30 seconds

Generate an overdue report automatically

Store at least 5,000 book records persistently

Produce a monthly usage statistics report

Complex Techniques:

File I/O with CSV structured data handling

Binary search algorithm for efficient book lookup

Object-oriented design (Book, Member, Loan classes)

Part 4: Common Pitfalls and How to Avoid Them

Pitfall	Why It Fails	How to Avoid
Fake client	IB moderators can detect invented clients	Choose a real person; document all contact
Trivial problem	Insufficient complexity for Criterion C	Ensure problem requires multiple features and complex techniques
No client consultation evidence	Criteria A and D require proof of interaction	Save all emails, meeting notes, photos
Vague success criteria	Cannot be tested or evaluated	Use numbers, times, quantities
Video too long or unclear	Moderator cannot verify functionality	Script the video; keep to 7 minutes; show each criterion clearly
Design after coding	Design must precede development	Complete design before writing code
Plagiarised code	Academic misconduct; zero marks	Write your own code; cite any tutorials used
Missing video evidence	Criterion C cannot be verified	Record, upload, and test link before submission

Part 5: EAL Support for the IA

Challenge	Support Provided
Writing in academic English	Bilingual templates; sentence starters; peer review with fluent speakers
Understanding command terms	Bilingual glossary of IA-specific terms; exemplar annotations
Client communication	Permission to conduct consultations in L1; translation of key documents permitted
Technical vocabulary	Bilingual code commentary template; pre-translated common phrases

Part 6: Timeline and Checklists

Final Submission Checklist

- ☐ Criterion A: Planning (client, problem, rationale, success criteria, record of tasks)
- ☐ Criterion B: Design (record of tasks, system overview, algorithms, test plan)
- ☐ Criterion C: Development (complex techniques, commented code, video evidence)
- ☐ Criterion D: Evaluation (testing evidence, client feedback, future improvements)
- ☐ Word count reasonable (no maximum, but quality over quantity)
- ☐ Video uploaded and link working
- ☐ File naming follows IB requirements
- ☐ Coversheet completed with candidate details
- ☐ Teacher has reviewed draft and given feedback
- ☐ No plagiarism; all sources cited
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